

# An Unrealised Isolated Type over $\ell_1$

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2026-06-20

## Source question

Ben Yaacov and Henson ask in Question 6.8 of [1]:

Is there any infinite-dimensional  $E$  other than  $\mathbb{G}$  over which the isolated types are dense? Is there any infinite-dimensional  $E$  over which there are unrealised isolated types? Specifically, what happens in the case  $E = \ell_1$ ,  $E^* = \ell_\infty$ , to which Proposition 6.7 does not apply?

and therefore (by continuity of  $f^*$ ) at some non-boundary point. By Lemma 4.14,  $f$  is realised. ■

Over  $E = \mathbb{G}$  the isolated types are again exactly the realised ones, but in this specific case they are dense and  $\mathbb{G}[E] = E$ .

*Question 6.8.* Is there any infinite-dimensional  $E$  other than  $\mathbb{G}$  over which the isolated types are dense? Is there any infinite-dimensional  $E$  over which there are unrealised isolated types? Specifically, what happens in the case  $E = \ell_1$ ,  $E^* = \ell_\infty$ , to which Proposition 6.7 does not apply?

## 7. COUNTING TYPES

We conclude with a calculation of the size of the type-space over a separable Banach space  $E$ . By “size” we mean here the metric density character (since the cardinal  $|S_n(E)|$  is the continuum as soon as  $n > 0$  and  $E \neq 0$ ).

**Theorem 7.1.** *Let  $E$  be a separable Banach space.*

This packet answers the second question positively, and does so in the specific case  $E = \ell_1$ . It does not settle whether isolated types are dense over  $\ell_1$ .

## Background from the source paper

For a Banach space  $E$ , one-types over  $E$  are identified in [1] with convex Katetov functions  $f \in K_C(E)$ . If  $f^*$  is the Legendre-Fenchel conjugate, then Lemma 4.12 of [1] says that a closed convex function  $h$  on  $B(E^*)$ , extended by  $+\infty$  outside  $B(E^*)$ , is the conjugate of some  $f \in K_C(E)$  provided

$$h(\lambda) + h(-\lambda) \leq 0 \quad (\lambda \in B(E^*)).$$

Lemma 5.2 says that  $f$  is local exactly when  $f^*$  is continuous on  $B(E^*)$ , and Theorem 5.7 says that  $f$  is isolated exactly when it is both local and  $\partial$ -extreme. Finally, Lemma 4.14 says that a type is realised in  $E$  exactly when  $f^*(0) = 0$ .

## Proof intuition

The weak-star unit ball of  $(\ell_1)^* = \ell_\infty$  is the product cube  $[-1, 1]^\mathbb{N}$ , whose extreme points are the sign sequences  $\{-1, 1\}^\mathbb{N}$ . Put the odd parity value

$$p(\varepsilon) = \varepsilon_1 \varepsilon_2 \varepsilon_3$$

on those extreme points, and take its convex envelope  $h$  on the whole cube. The oddness of  $p$  gives the antipode inequality. The fact that  $h$  is the convex envelope of its extreme-point values makes it impossible to raise  $h$  on the boundary while preserving the antipode inequality, so the corresponding type is  $\partial$ -extreme. Since this envelope depends on only three coordinates, it is weak-star continuous. At the origin, however, the four sign vectors of parity  $-1$  average to 0, so  $h(0) = -1$ ; hence the type is not realised.

## The construction

Let

$$K = [-1, 1]^\mathbb{N}$$

with the product topology, identified with  $B(\ell_\infty)$  in the weak-star topology  $\sigma(\ell_\infty, \ell_1)$ . Let  $\Omega = \{-1, 1\}^\mathbb{N}$ , the set of extreme points of  $K$ , and define

$$p(\varepsilon) = \varepsilon_1 \varepsilon_2 \varepsilon_3 \quad (\varepsilon \in \Omega).$$

For  $\lambda \in K$ , set

$$h(\lambda) = \inf \left\{ \int_\Omega p(\varepsilon) d\mu(\varepsilon) : \mu \in \mathcal{P}(\Omega), \int_\Omega \varepsilon_n d\mu(\varepsilon) = \lambda_n \text{ for every } n \right\}.$$

The set of representing measures is never empty; for example, take the product measure with one-dimensional means  $\lambda_n$ . Since  $p$  only depends on the first three coordinates,  $h$  is just the convex envelope on the three-dimensional cube of  $(s_1, s_2, s_3) \mapsto s_1 s_2 s_3$  on  $\{-1, 1\}^3$ . In particular  $h$  is a finite continuous convex function on  $K$ , depending only on  $\lambda_1, \lambda_2, \lambda_3$ .

**Lemma 1.** *The function  $h$  satisfies  $h(\lambda) + h(-\lambda) \leq 0$  for all  $\lambda \in K$ .*

*Proof.* Let  $\mu$  represent  $\lambda$ , and let  $\check{\mu}$  be its image under  $\varepsilon \mapsto -\varepsilon$ . Then  $\check{\mu}$  represents  $-\lambda$ , while

$$\int p d\check{\mu} = \int p(-\varepsilon) d\mu(\varepsilon) = - \int p d\mu,$$

because  $p$  is odd. Therefore  $h(-\lambda) \leq - \int p d\mu$ . Taking the infimum over all representing  $\mu$  gives  $h(\lambda) \leq -h(-\lambda)$ .  $\square$

By Lemma 1 and the conjugate criterion of [1, Lemma 4.12], the function

$$f(x) = \sup_{\lambda \in K} (\lambda(x) - h(\lambda)) \quad (x \in \ell_1)$$

belongs to  $K_C(\ell_1)$ , and  $f^* = h$ . Thus  $f$  is a one-type over  $\ell_1$ .

**Lemma 2.** *The type  $f$  is  $\partial$ -extreme.*

*Proof.* Let  $g \in K_C(\ell_1)$  satisfy  $g \leq f$ , and put  $k = g^*$ . Then  $k \geq h$  on  $K$ , and  $k$  satisfies the same antipode inequality

$$k(\lambda) + k(-\lambda) \leq 0.$$

We first show that  $k = h$  on the extreme points  $\Omega$ . If  $\varepsilon \in \Omega$ , then

$$k(\varepsilon) \leq -k(-\varepsilon) \leq -h(-\varepsilon) = h(\varepsilon),$$

because  $p(-\varepsilon) = -p(\varepsilon)$  and  $h$  agrees with  $p$  on  $\Omega$ . Together with  $k \geq h$ , this gives  $k(\varepsilon) = h(\varepsilon)$  for all  $\varepsilon \in \Omega$ .

Now fix  $\lambda \in K$ . For every representing measure  $\mu$  on  $\Omega$ , Jensen's inequality for the lower semicontinuous convex function  $k$  gives

$$k(\lambda) \leq \int_{\Omega} k(\varepsilon) d\mu(\varepsilon) = \int_{\Omega} h(\varepsilon) d\mu(\varepsilon) = \int_{\Omega} p(\varepsilon) d\mu(\varepsilon).$$

Taking the infimum over all representing measures yields  $k(\lambda) \leq h(\lambda)$ . Since  $k \geq h$ , we have  $k = h$  on all of  $K$ , in particular on  $\partial B(\ell_{\infty})$ . This is exactly  $\partial$ -extremality of  $f$ .  $\square$

**Theorem 1.** *There is an isolated one-type over  $\ell_1$  that is not realised in  $\ell_1$ .*

*Proof.* The conjugate  $f^* = h$  is continuous on  $B(\ell_{\infty})$ , so  $f$  is local by [1, Lemma 5.2]. By Lemma 2,  $f$  is also  $\partial$ -extreme. Therefore  $f$  is isolated by [1, Theorem 5.7].

It remains to see that  $f$  is not realised. Let  $A \subset \{-1, 1\}^3$  be the four sign triples with product  $-1$ . Their average is  $(0, 0, 0)$ , so the uniform measure on  $A$ , extended by any product measure on the remaining coordinates with mean zero, represents  $0 \in K$  and has  $p$ -integral  $-1$ . Since  $p \geq -1$ , this proves  $h(0) = -1$ .

By [1, Lemma 4.14], a one-type over  $E$  is realised in  $E$  if and only if its conjugate takes value 0 at the origin. Since  $f^*(0) = h(0) = -1$ , the isolated type  $f$  is not realised in  $\ell_1$ .  $\square$

## Verification and limitations

The proof is purely analytic. The script `code/parity_envelope_check.py` verifies the finite parity fact used at the origin: the four parity- $-1$  vertices in  $\{-1, 1\}^3$  average to  $(0, 0, 0)$ , and the parity integral there is  $-1$ .

This is a full positive answer to the unrealised-isolated-type subquestion in Question 6.8, including the specific case  $E = \ell_1$ . It does not prove that the isolated types over  $\ell_1$  are dense, and it does not characterise all separable Banach spaces  $E$  for which  $\mathbb{G}[E]$  exists.

## Novelty check

The run indexes were checked for arXiv:1211.4814, the paper title, generic orbits, type isolation, Gurarij/Gurarii terms, prime models,  $G[E]$ , and  $\ell_1$ . No exact prior packet was present. A bounded web search for the source title and the phrases ‘‘Gurarij isolated types  $\ell_1$ ’’, ‘‘ $G[E]$  Gurarij  $\ell_1$ ’’, and ‘‘unrealised isolated types Gurarij’’ found the source paper but no obvious later answer. Local later citation hits to [1] were also searched for the relevant keywords and did not reveal a closure of Question 6.8.

## Human review recommendation

Review as a candidate full solution to the second part of Question 6.8. The main point to check is Lemma 2: the convex-envelope maximality on the extreme sign sequences is what upgrades the continuous parity envelope from a convex Katetov type to an isolated type.

## References

- [1] I. Ben Yaacov and C. W. Henson, *Generic orbits and type isolation in the Gurarij space*, Fundamenta Mathematicae 237 (2017), 47–82. arXiv:1211.4814.